MATH 128a: Homework #2

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Due by: **Wednesday, September 17, 2025, 11:59pm**

Problem 1: Problems from textbook Section 1.2

Solve problems 1a, 2d(1h), 3b, 6a(5a), 12ab(10ab), 19abc(15abc), 20a (16a) from Section 1.2 of the textbook.
If the problem has a different number in edition 9 of the textbook, it is in parentheses.
For problem 20a: write the result in the form $d.ddddddddd + 2^{-dd}$, where d is a decimal digit, to avoid a ridiculously long answer.

Answer of Problem 1

1. Compute the absolute error and relative error in approximations of p by p^*.

   a. $p = \pi$, $p^* = 22/7$

   Solution: The absolute error is $|\pi - 22/7| \approx 0.00126448926735$, and the relative error is $\frac{|\pi - 22/7|}{|\pi|} \approx 0.0004024994347707$.

2. Compute the absolute error and relative error in approximations of p by p^*.

   d(1h). $p = 9!$, $p^* = \sqrt{18\pi}(9/e)^9$.

   Solution: The absolute error is $|362880 - 359536.87| = 3343.13$. The relative error is $|362880 - 359536.87|/9! = 0.009213$.

3. Suppose p^* must approximate p with relative error at most 10^{-3}. Find the largest interval in which p^* must lie for each value of p.

   b. 900

   Solution: We need to find an interval $[a, b]$ such that $\frac{|p^* - 900|}{|900|} < 10^{-3}$ for every $p^* \in [a, b]$. So we solve this equation at the end points $\frac{|p^* - 900|}{900} = 10^{-3}$ which yields the solutions $p_{max} = 900.9$ and $p_{min} = 899.1$. So the interval is $[899.1, 900.9]$.

6(5). Use three-digit rounding arithmetic to perform the following calculations. Compute the absolute error and relative error with the exact value determined to at least five digits.

a. 133+0.921

Solution: Adding gives 133.921 and rounding to three places gives 134. The absolute error is 0.079 and the relative error is $\frac{0.079}{133.921} \approx 0.000590$

12(10). The number e can be defined by $e = \sum_{n=0}^{\infty}(1/n!)$. Compute the absolute error and relative error in the following approximations of e.

a. $\sum_{n=0}^{5} \frac{1}{n!}$

Solution: $\sum_{n=0}^{5} \frac{1}{n!} = 1 + 1 + 1/2 + 1/6 + 1/24 + 1/120 = \frac{326}{120} = 2.716667$ Using a calculator's approximation of $e = 2.71828182846...$, the absolute error is approximately 0.001615 and the relative error is approximately 5.94×10^{-4}.

b. $\sum_{n=0}^{10} \frac{1}{n!}$

Solution: $\sum_{n=0}^{10} \frac{1}{n!} = 1 + 1 + 1/2 + 1/6 + 1/24 + 1/60 + 1/720 + 1/5040 + 1/40320 + 1/362880 + 1/3628800 = \frac{9864101}{3628800} = 2.71828180115$. The absolute error is approximately 0.00000002731 and the relative error is approximately 0.0000000100

19(15). Use the 64-bit long real format to find the decimal equivalent of the following floating-point machine numbers.

a. 0 10000001010 10010011000000000000000 00000000000000000000000000

Solution: We have $s = 0$, $c = 2^{10} + 2^3 + 2^1 = 1034$ and $f = 1/2^1 + 1/2^4 + 1/2^7 + 1/2^8 = 0.57421875$. So the number is $(-1)^0 \cdot 2^{1034-1023} \cdot (1.57421875) = 3224$

b. 1 10000001010 10010011000000000000000 00000000000000000000000000

Solution: This is the same as part a. except for the sign bit, so the number is -3224.

c. 0 01111111111 01010011000000000000000 00000000000000000000000000

Solution: We have $s = 0$, $c = 2^{10} - 1 = 1023$ and $f = 1/2^2 + 1/2^4 + 1/2^7 + 1/2^8 = 0.32421875$. So the number is $(-1)^0 \cdot 2^{1023-1023} \cdot (1.32421875) = 1.32421875$

20(16). Find the next largest and smallest machine numbers in decimal form for the numbers given in Exercise 19 (15).

a. 0 10000001010 10010011000000000000000 00000000000000000000000000

Solution: The next larger number in binary is obtained by adding a 1 in the last bit of the mantissa, i.e.
0 10000001010 10010011000000000000000 00000000000000000000000001,
thus we have $s = 0$, $c = 2^{10}+2^3+2^1 = 1034$ and $f = 1/2^1+1/2^4+1/2^7+1/2^8+1/2^(52) = 0.57421875+2^{-52}$. So the next larger number is $(-1)^0 \cdot 2^{1034-1023} \cdot (1.57421875+2^{-52}) = 3224 + 2^{-41}$. For the next smaller number, we need to decrease the mantissa in the last digit. So we obtain
0 10000001010 10010010111111111111111 11111111111111111111111111,
thus we have $s = 0$, $c = 2^{10} + 2^3 + 2^1 = 1034$ and $f = 1/2^1 + 1/2^4 + 1/2^7 + \sum_{j=9}^{52} 1/2^j = 0.57421875-2^{-52}$. So the next smaller number is $(-1)^0 \cdot 2^{1034-1023} \cdot (1.57421875-2^{-52}) = 3224 - 2^{-41}$.

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